

While there are many different technologies of interest in the storage industry, there are some nascent areas too such as computational storage. Existing technologies are being standardised, and there is renewed interest in standardised storage fabric management.

Cloud architecture: The larger vendors use their own storage technologies most of the time. But the usage pattern, be it block, object or file storage, depends on the convenience of use. If the priority is performance or cost, technologies like object and file are appropriate. “Sometimes clients prefer data locality -- local storage and that particular region support kind of storage,” said Krishna Kumar M., technology architect cloud at Accenture, speaking at the event.

“The technologies are being offered by the big cloud providers. And it is the convenience of usage and the business needs that decide what kind of storage we use. This includes cloud native technologies and similar technologies in combination,” he added.

Cloud storage is being mainly used for disaster recovery (DR) and backup. The cloud is heavily used for backing up the data set, which is also backed up in some other region for DR. Kumar pointed to the US model in which data is stored in the western, central and eastern regions to ensure it survives catastrophic incidents. “An increasing number of people are taking backup of additional data sets to meet their RPO (recovery point objective) and RTO (recovery time objective),” Kumar said.

File management is another common trend, which aims at high usability and convenient sharing of files.

Companies like Linbit are actively encouraging users to build their private clouds. Philipp Reisner, CEO of Linbit, said that companies are combining on-premises clouds or private clouds and public clouds. They prefer locating their data in a place that is accessible and secure, at an economical cost.

Blockchain storage

ActionSpot is a company that focuses on blockchain tools. “We also use a lot of Kubernetes. It will probably be pretty dominant in the next ten years. Blockchain is a part of cryptography and data security,” said Olga Buchonina, CEO and founder, ActionSpot Corp.

The blockchain is on the cloud and has never used onsite infrastructure. “The FinTech sector, such as banks like HSBC, is adopting blockchain technology,” said Buchonina.

She added, “This just indicates that we are moving forward. However, in some cases, having hybrid solutions makes more logical sense than having a pure cloud, especially if it’s a security concern to have an on-premises solution. So people need to understand what’s needed. Blockchain technology uses cloud, Dockers, Kubernetes and container based databases.”

Stop obsessing on storage type

Alex McDonald, director, SNIA EMEA, encouraged everybody to stop obsessing about the difference between NVMe or small computer system interface (SCSI), unless the developers are deeply involved with the storage hardware level or a certain layer above that. “Stop obsessing about the difference between file block and object,” he said.

He explained further, pointing to an instance that used POSIX semantics for doing file open, read and write clauses. “That is the way we need to think about all data access,” he said. For example, if there is a replication issue, the developer should stop replicating at the wrong level. And the wrong level may be, for instance, in the NoSQL database that you choose, while you let it replicate. “Replication shouldn’t know anything about the underlying storage architecture, its frailties, its fragility or lack of performance or whatever it might be that needs to be

dealt with in a layer below, where we need to get the levels of abstraction,” McDonald added.

It is wiser to focus on the application with knowledge about what data it uses, what data is really needed from the application consumption perspective, and make sure the data services are abstracted to virtually where all the workloads can benefit.

Edge acceleration

According to Reisner, edge deployment is on the rise but there is a need for automation. The IoT devices connected at the network edge are being pushed to take on more storage functions. They can process data in small quantities before sending it to a central storage location. The raw data can be processed or managed at the edge.

Edge-specific use cases is another area the panelists in the discussion thought may need a lot of work, specifically, having the data replicated from the edge to the cloud backend, and intercloud replications.

Emerging technologies and use cases

Kumar also talked of access control. “When there is a data set, it has to access the General Data Protection Regulation (GDPR) compliance and other requirements, maybe row level or column level data. Only a small part of a data set may need to be accessed. Hence, we need specific access mechanisms,” he said.

Second, if the user is able to access the data in the US, he or she may not be able to access it when travelling to some other country, because the data is restricted to that specific region. “So we are seeing a lot of access mechanisms and technologies in place where the data is not exposed beyond certain boundaries,” Kumar explained. He said this will bring in more stringent technologies that allow data to be granularised and accessed.